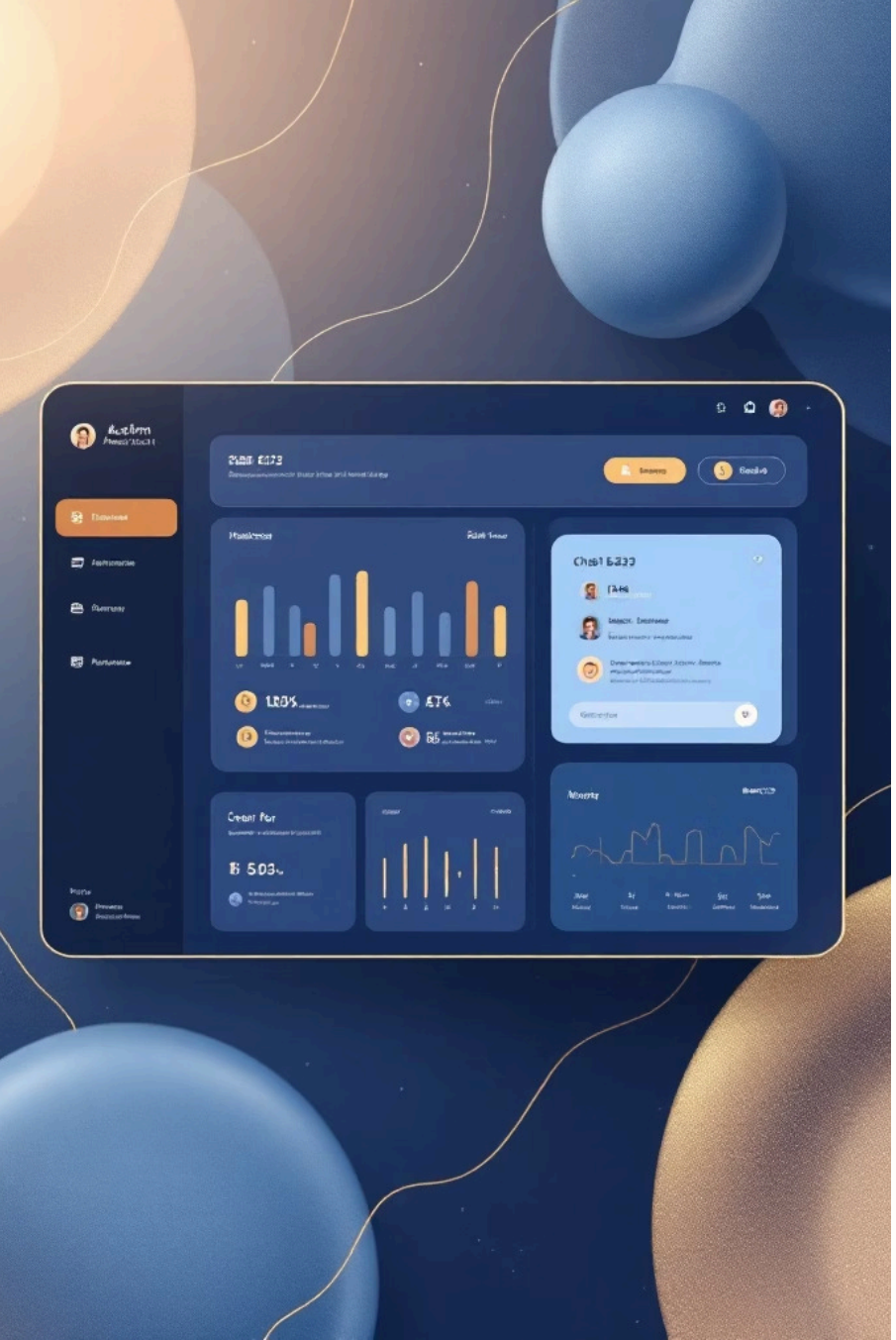
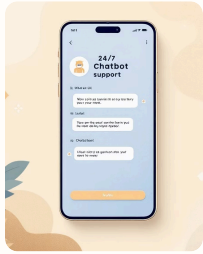


WellBot — AI Wellness Chatbot System (Milestone 2)

Group presentation • Modern healthcare AI theme • Concise technical summary for stakeholders and reviewers

presented by Team-A

Project Overview

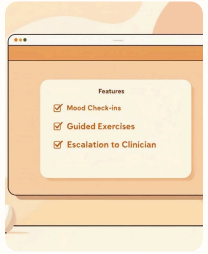


Always-on Wellness

Continuous, low-friction access to mental health resources and conversational support.

Security-first Design

Data encryption, token-based auth, and privacy-preserving storage for sensitive user data.



Core Functionality

Symptom triage, mood detection, guided CBT-style micro-interventions, escalation paths.

Natural Language Processing (NLP)

Text Understanding

Tokenization, entity extraction, medical and emotional phrase vocabularies tuned for mental-health context.

Sentiment Analysis

Continuous sentiment scoring to detect distress, negative affect, and recovery trends over time.

Intent Recognition

Classifiers for help-seeking, info requests, crisis signals, and appointment intent with calibrated thresholds.

Mood Detection

Multimodal cues (text, quick emojis) produce a probabilistic mood state used to adapt responses.

Machine Learning (ML)

- Adaptive models learn from anonymized interaction logs to improve response relevance.
- Personalization layers tailor phrasing, recommended exercises, and pacing to user profiles.
- Predictive models identify risk trajectories and suggest proactive outreach.
- Continuous evaluation with holdout sets and A/B experiments for safe rollout.

Large Language Models (LLM)

- ## Human-like Responses

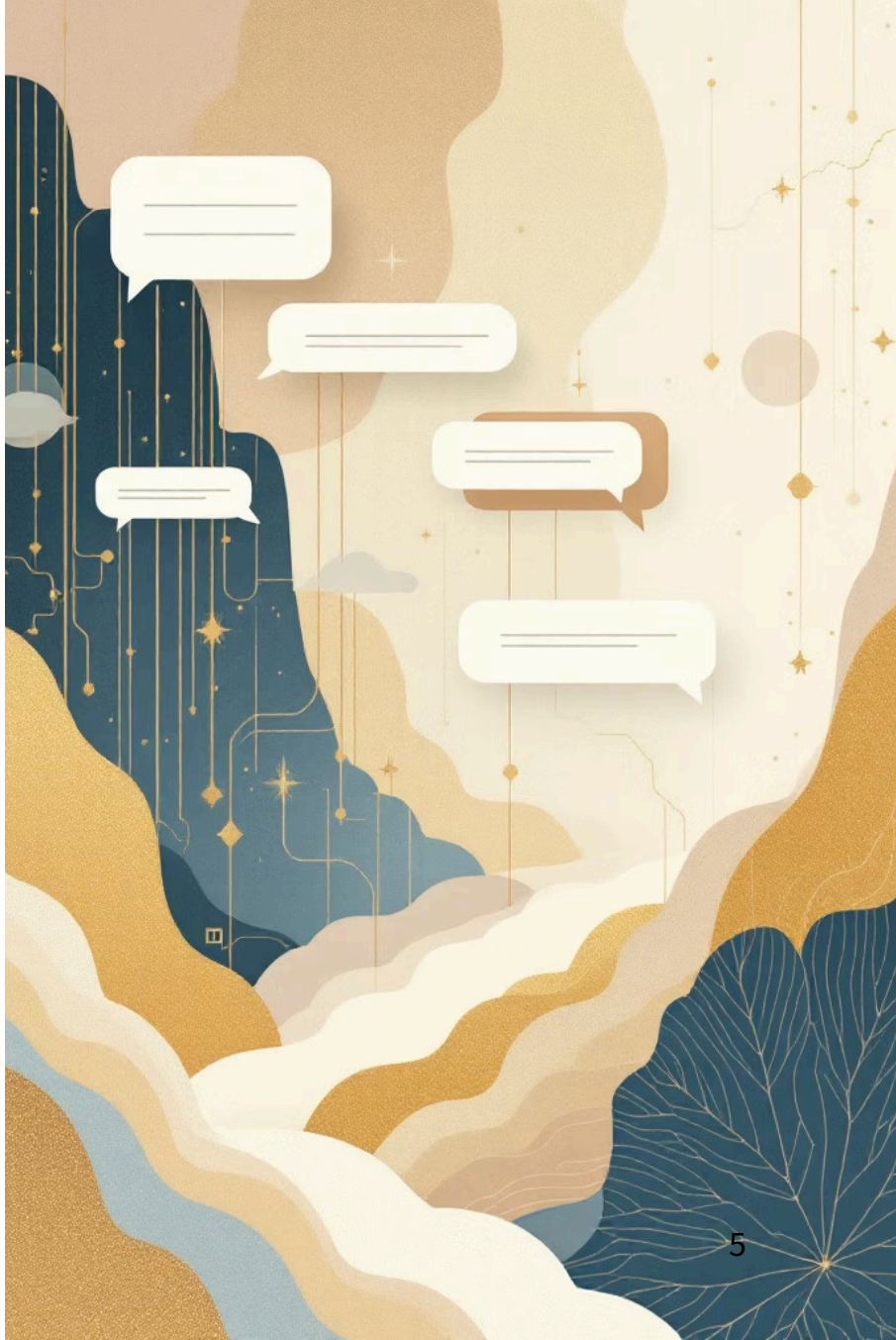
  Generative models provide fluent, context-aware dialogue aligned with clinical tone constraints.

- ## Contextual Understanding

  Session-level context windows + short-term memory to maintain coherence and therapeutic continuity.

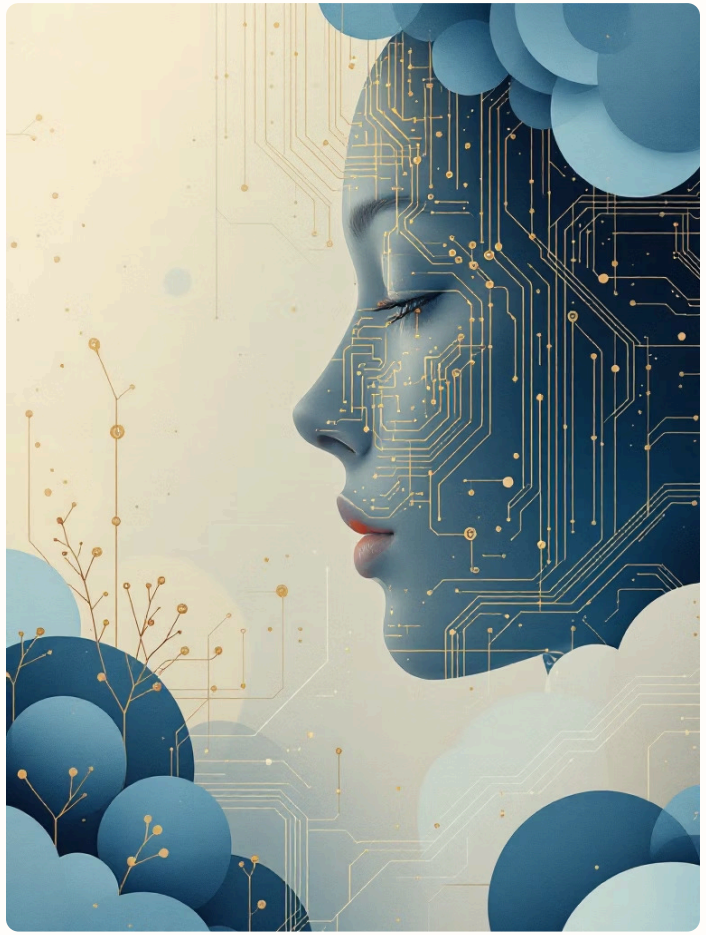
- ## Safety Layers

  Response filters and clinician-curated templates reduce hallucinations and ensure evidence-based guidance.



Chatbot System Architecture

Diagram shows core pipeline: capture, analyse, decide, generate, log. Each stage includes monitoring and privacy controls.



AI Architecture — Hybrid Approach

01

Rule-based Layer

Deterministic safety checks, crisis escalation rules, and compliance enforcement.

02

Generative Layer

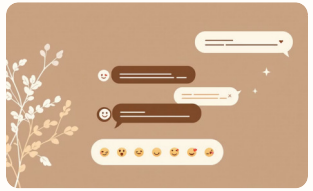
LLM-driven natural language generation constrained by templates and safety filters.

03

Orchestration

Decision engine chooses between layers based on confidence, risk score, and policy.

Dashboard Interface



Chat UI

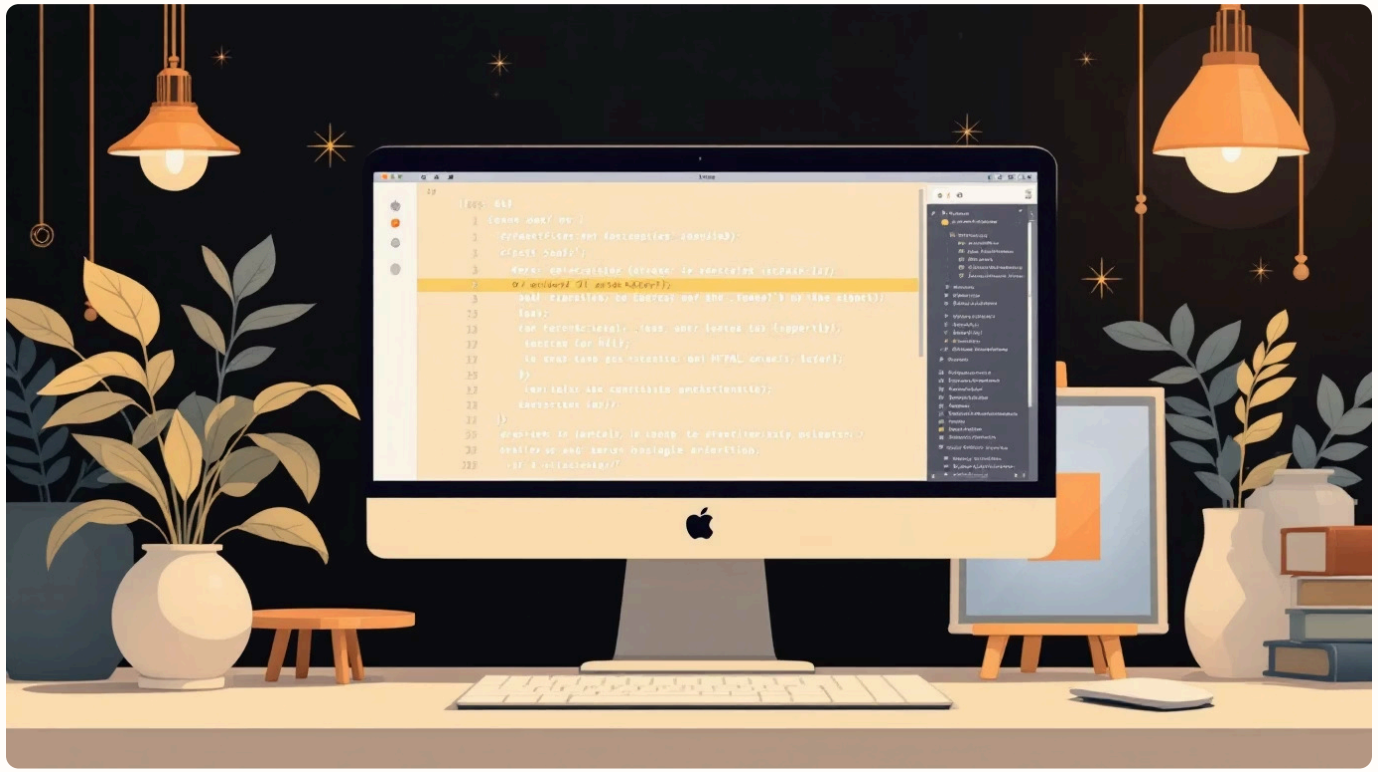
Minimal, accessible chat with suggested micro-responses and clear escalation buttons.

Mood Selection

Quick mood check-ins with trend view to inform personalization and clinician handoff.

User Interaction

Session summaries, timestamps, and clinician notes integrated for oversight.



Frontend Technologies

- HTML & semantic structure for accessibility and content clarity.
- CSS with glassmorphism patterns and theme variables (primary accent #B88E23).
- JavaScript for interactive chat, state management, and client-side validation.
- Responsive design: mobile-first, ARIA attributes, keyboard navigation.

Backend, Database & Security

Backend — Flask API

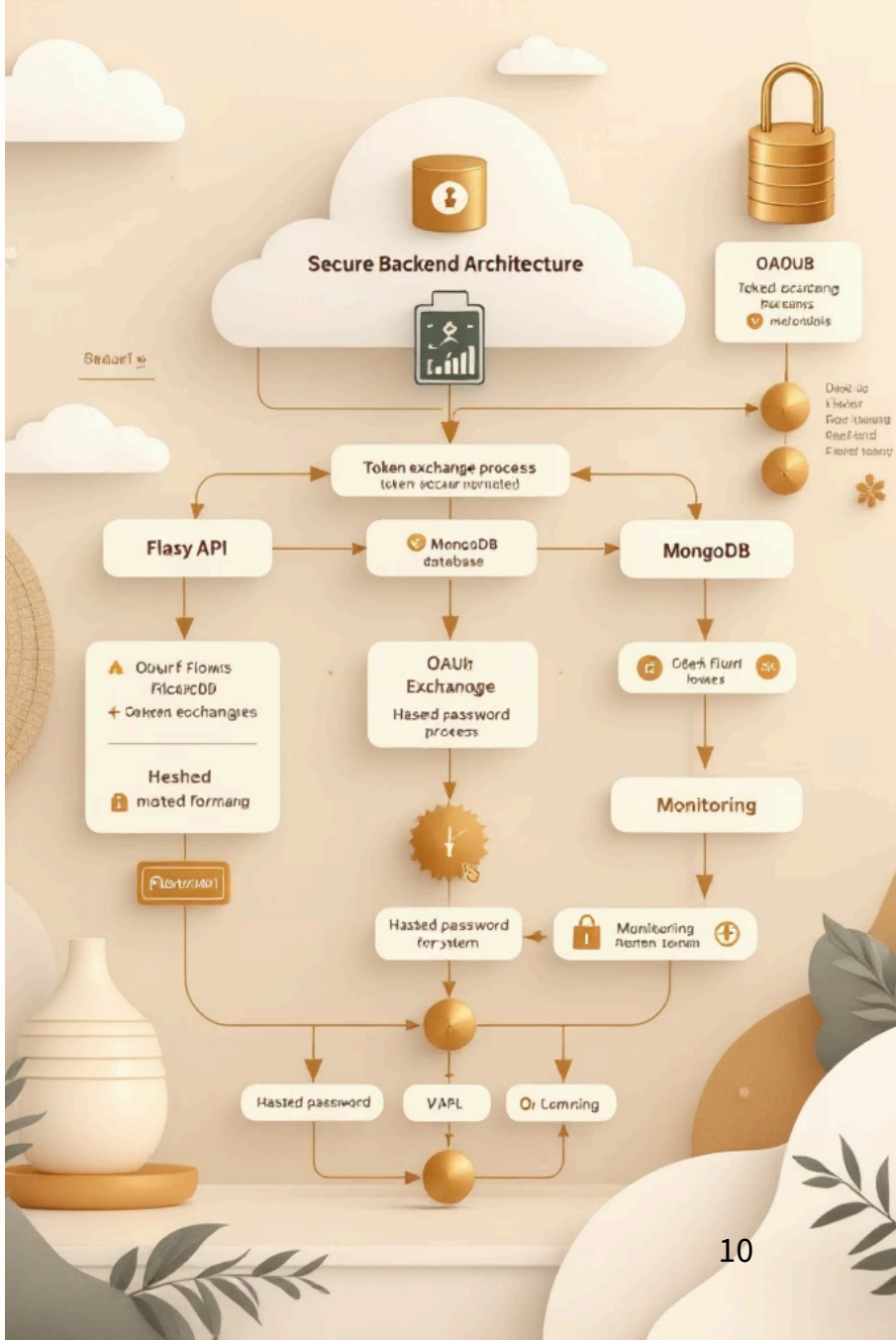
Lightweight REST endpoints, request validation, rate limiting, and logging for reproducible workflows.

Database — MongoDB

Document storage for session transcripts, metadata, and de-identified analytics with access controls.

Security & Auth

Google OAuth for user sign-in, SHA-256 hashing for sensitive tokens, JWT token authentication and rotation.

Testing, Validation & Next Steps

Unit and integration tests, simulated user studies, safety validation, and roadmap: pilot, clinician integration, regulatory review.